

**Centrality Claims in Computer Science Research:
A Rhetorical Feature for Establishing Significance**

Sungjun Lee and Wendi Liu

University of British Columbia

WRDS 150B: Writing and Research in the Disciplines

Dr. Susan Blake

August 12, 2024

[Abstract]

[Keywords]

Introduction

Swales (1990) developed the Create-a-Research-Space (CARS) model for the main rhetorical moves found in research article (RA) introductions. Within this model, “Claiming Centrality” is a possible step of Move 1 (Swales 1990). Centrality claims are crucial for establishing the topic’s significance and position within academic discourse, and they are often present in the introductions of computer science RAs. Additional research on the CARS model shows that centrality claims can be classified into two categories: research and real-world, which stresses importance in the theoretical research world and emphasizes practicality in the real world, respectively (Samraj, 2002).

Though previous studies have been conducted on the general topic of the CARS model in RA introductions (Cortes, 2013; Samraj, 2002), few focus on centrality claims. Furthermore, there is little research about the varying functions of research versus real-world centrality claims. Knowledge on the feature of “Claiming Centrality” is especially limited in computer science, relative to other disciplines.

Our study aims to investigate the research question: *How are the two different types of centrality claims used in computer science research article introductions?* The goal is to analyze the centrality claims in a corpus of computer science RAs and determine any variations in the rhetorical usage and function of the two different types. We examine 24 computer science RAs from two publications: *International Journal of Information Management* and *Journal for Machine Learning Research*. Using a qualitative approach, we identify the centrality claims in the RA introductions and classify them into research and real-world claims.

Based on initial observations, centrality claims are often found at the beginning of the introductory paragraphs, so we expect them to have a prominent role in setting the foreground of computer science RAs. Both types of centrality claims appear frequently in our corpus, and preliminary data suggests that their usage is related to the RA's topic. Thus, investigating computer science centrality claims can also offer insights about the prevalence of certain research areas within the field.

This study furthers our understanding of the discourse feature of centrality claims in the genre of research articles. It can reveal new knowledge on the various functions of centrality claims, expanding on Swales' (1990) original definition and Samraj's (2002) addition of the different classifications. Our investigation can particularly discover more about the rhetorical function and usage of centrality claims in computer science research, which has not been extensively studied previously. These findings can also have pedagogical benefits for instructors and students of academic writing, and improve the effectiveness of presenting research.

The remainder of our research article is outlined as follows: a description of the process for constructing the corpus and the methodology for collecting data on centrality claims, then a discussion of results detailing the data and its analysis, and finally, a presentation of the conclusions and impacts of this study.

Methods

Corpus Construction

The combined corpus includes 24 computer science RAs, totaling approximately 275,000 words, drawn from two prominent peer-reviewed journals: the *International Journal of Information Management (IJIM)* and the *Journal for Machine Learning Research (JMLR)*. All

articles are from the same discipline of computer science to maintain a singular thematic focus for the corpus, which is recommended by corpus linguists (Bauer & Aarts, 2000).

Both journals were selected based on their high impact factors and relevance to current research trends in computer science. The *IJIM* was selected because of its comprehensive coverage of developments in information management, a critical area in artificial intelligence. Conversely, the *JMLR* focuses on cutting-edge research in machine learning, a subfield in the broader discipline of artificial intelligence. It was chosen based on its high ranking in Google Scholar Metrics (Artificial Intelligence, 2024) and its frequent appearance in the publications of faculty members specializing in artificial intelligence at the University of British Columbia.

For the corpus constructed by Sungjun Lee (see Appendix A), the *IJIM* was chosen as the primary source, covering the most recent publication years of 2022 to 2024. A total of twelve articles, approximately 120,000 words, were selected from the first issue of each volume within this time frame to ensure coverage of the most recent research trends and methodologies. These articles encompass diverse topics such as artificial intelligence, design innovation, machine learning, and cybersecurity. This selection method follows Becher's (1994) insight on representing various subfields within a discipline to enhance the validity of research analysis.

Wendi Liu's corpus (see Appendix B) contains approximately 155,000 words, consisting of twelve articles from the *JMLR* selected from the same publication years (2022-2024). Articles being taken from the same date range ensures synchronicity, allowing for the identification of patterns within the corpus (Bauer & Aarts, 2000). Since the *JMLR* is not a periodical, but rather publishes each article immediately upon finalization, each year is one volume, and each article belongs to its own issue (Journal for Machine Learning Research, 2024). Four issues were selected arbitrarily from each volume for a total of twelve articles. This random selection

process, as recommended by corpus linguists (Hyland, 2011), ensures a broad representation of the journal's content.

The combined corpus of 24 RAs from two high-impact journals covers a wide range of topics in computer science. This diverse yet focused corpus provides a dataset for analyzing the discourse move of "Claiming Centrality" in the introductions of computer science RAs.

Data Collection

While making observations about rhetorical features in our combined corpus' RAs, the first move from Swales' (1990) CARS model was most notable. Though Move 1 contains three steps (Swales, 1990), a previous study shows that it can be challenging to provide distinct definitions for and clearly differentiate between Step 2's topic generalizations and Step 3's literature reviews (Samraj, 2002). The initial coding and tagging of our corpus encountered this issue as well. Therefore, our research team eventually decided to focus solely on the discourse feature of Step 1: "Claiming Centrality" within our corpus of computer science RAs.

Swales (1990) defines centrality claims as "appeals to the discourse community whereby members are asked to accept that the research about to be reported is part of a lively, significant or well-established research area" (Swales, 1990, p. 144). Authors use centrality claims to establish the importance and relevance of their work. The rhetorical feature aims to show that the general topic of the specific study is central to current research. This operational definition guided the tagging of centrality claims in our corpus.

We identified instances of centrality claims based on language that emphasizes significance. As a preliminary scan, we searched for keywords such as "important", "extensive", and "prevalent" using Adobe Acrobat Reader's software. With a relatively small corpus, we could then read each RA's introduction section to find centrality claims that we missed during the

initial text search. Each occurrence of the feature was tagged by highlighting the phrases that pertain to the centrality claim.

However, we excluded instances that contained the type of language in centrality claims but did not have a purpose of demonstrating the research area's significance. For example, a sentence with language that stresses importance but serves to indicate a gap in knowledge, or one that uses similar keywords but not in reference to the article's main topic, would not qualify as centrality claims. Data reduction is a crucial step in methodology (Smagorinsky, 2008), and corpus linguists employ it to ensure the relevance of their data (e.g., Hyland, 1999). By our operational definition, these cases are not directly related to the authors' claims of the research's centrality, and would not be considered relevant to our study.

Finally, we classified the centrality claims in our RAs, according to the categories of centrality claims identified by Samraj (2002): research and real-world. We coded these two different types to observe patterns within our corpus about the domain in which they claim centrality — either the theoretical research world or the pragmatic real world. This allows for further analysis of the discourse feature of centrality claims in the context of articles from high-impact journals in the field of computer science.

Results and Discussion

Centrality claims were present in all but one of the 24 articles in our corpus. It is shown to be a prominent feature that consistently appears in the introductions of our computer science RAs. We found that most RAs contained both types of centrality claims, while several only used research claims, and only one article solely used real-world claims. The data reveals that research centrality claims are used in almost every RA of our corpus. Table 1 contains the quantitative data on the articles that contain each type of centrality claims.

Table 1*Number of Articles Containing Different Types of Centrality Claims*

Type	Research Only		Real-World Only		Both	
Journal	<i>IJIM</i>	<i>JMLR</i>	<i>IJIM</i>	<i>JMLR</i>	<i>IJIM</i>	<i>JMLR</i>
Number By Journal	2	4	1	0	9	7
Number By Type	6		1		16	
Total	12 (<i>IJIM</i>) + 11 (<i>JMLR</i>) = 23					

The table shows that 22 of the RAs claim centrality in research or both domains. It appears that computer science writers tend to stress the importance of their topic in the research world. Only one article that contains real-world claims does not have the presence of any research claims.

The vast majority of real-world claims appearing alongside research claims shows that the practical applications are often connected with the theoretical side of computer science research.

By the corpus data, both types of centrality claims appeared frequently. Nevertheless, we identified more cases of research centrality claims. In Table 2, we quantitatively summarize each occurrence of centrality claims in our corpus. Multiple centrality claims within one sentence were counted separately, with the specific word or phrase that claims centrality as one instance.

Table 2*Instances of Centrality Claims in the Corpus*

Type	Research		Real-World	
Journal	<i>IJIM</i>	<i>JMLR</i>	<i>IJIM</i>	<i>JMLR</i>
Number By Journal	45	24	23	13
Number By Type	69		36	
Total	68 (<i>IJIM</i>) + 37 (<i>JMLR</i>) = 105			

Using our method for tagging, we found almost twice as many research centrality claims compared to real-world claims. This trend was apparent in both of the journals that formed our corpus. The result of having more research claims further supports the notion that authors for computer science RAs tend to focus on the theoretical significance of their research area when claiming centrality.

The Two Types of Centrality Claims

Research centrality claims are primarily used to position the research within the theoretical landscape of academic discourse. They emphasize the study's contribution to scholarly discussions in a broader field of knowledge. Their main rhetorical function is to establish the RA's significance by connecting it to fundamental theoretical concepts or current research advancements. These claims often use language that stresses the study's importance in relation to existing knowledge and its potential to deepen the present understanding.

The following excerpt features research centrality claims in the first few sentences in the RA's main body. The general research area of the RA is community detection.

- (1) Network is a *prevalent* form of relational data. A *central* theme in learning network data is community detection (Goldenberg et al., 2010; Fortunato, 2010). Community detection seeks to partition the nodes of a network into several disjoint subsets (a.k.a communities) upon observing the adjacency matrix (Girvan and Newman, 2002). The underlying assumption is that nodes within the same community share some commonalities in their connection patterns. To understand and to motivate algorithms for community detection, statisticians, probabilists and theoretical computer scientists have *studied stochastic blockmodels* (SBMs, Holland et al., 1983) *extensively*. To date, researchers have obtained a thorough understanding of the fundamental limits and the behavior of various algorithms under SBMs.

(Gao et al., 2022, p. 1 – emphasis added)

Terms such as “prevalent” and “central” highlight the significance of community detection in network data, an important form of relational data. After some elaboration about the topic itself, there is another centrality claim at the end of the excerpt, referring to extensive research activity in the algorithms of community detection. These are research centrality claims since they focus on the theoretical aspects of the research area.

On the other hand, real-world centrality claims focus on the practical implications and pragmatic applications of the research. They highlight the ways that the research area can solve real-world problems, improve practices, or impact industry standards. Their main rhetorical function is to demonstrate the research's relevance and utility in addressing immediate, real-world issues. These claims often use language that underscores the potential benefits and applications of the research in everyday life or specific industries.

The article by Bai et al. (2023) introduces its research by focusing on the practical applications of artificial intelligence in healthcare. The authors highlight the immediate benefits and real-world impact of their work, making it relevant to both practitioners and policymakers:

- (2) The healthcare industry is on the brink of a major transformation, driven by advances in artificial intelligence. Artificial intelligence (AI) applications in healthcare have the *potential to revolutionize patient care by improving diagnostic accuracy and treatment efficacy*. This research investigates the impact of AI on healthcare information processing, aiming to enhance stakeholder decision-making. Ultimately, our goal is to develop AI systems that can be seamlessly integrated into healthcare settings, providing tangible benefits to patients and providers alike.

(Bai et al., 2023, p. 3 — emphasis added)

Phrases like "revolutionize patient care" and "improving diagnostic accuracy" emphasize the practical significance of the research. Swales (1990) notes that establishing a territory through real-world centrality claims can appeal to the practical needs and interests of a broader audience.

This excerpt demonstrates how the research addresses practical challenges in healthcare, thus making it relevant to practitioners and policymakers. Samraj (2002) also discusses how real-world centrality claims can highlight utility to demonstrate the impact of research. Moreover, real-world claims expand the study's appeal beyond the academic community to include industry professionals.

As mentioned previously, RAs often employ both types of centrality claims instead of exclusively relying on one. The following excerpt is the entire first paragraph of the introduction of an article on the subject of multivariate extremes.

- (3) Multivariate extremes arise when one or more of rare extreme events occur simultaneously. They are of *paramount importance* for understanding environmental risks such as fires or droughts since they are driven by joint extremes of a number of meteorological variables. Similarly, catastrophic financial events are also of a multivariate nature in financial systems driven by core institutions that are connected. In the above examples one is precisely interested in modeling the dependence between rare individual extremes. Multivariate extreme value theory is an *active research area that provides tools for modeling such events*.

(Medina et al., 2024, p. 1 – emphasis added)

This excerpt first claims centrality in the practical applications of multivariate extremes, including environmental and financial disasters. Then, it uses another centrality claim to connect these real-world concerns to the theory, highlighted as an “active” area in the research world.

Overall, the choice for the type of centrality claim that is used appears to be linked to the RA's topic. If the article's research area concentrates on expanding an aspect of existing theory, it unsurprisingly tends to only contain research centrality claims, and not real-world ones. Real-world claims appear when the subject matter of the RA is also related to practical applications in some way. Nevertheless, this is not an absolute rule. There are instances where

real-world claims appear even when the main focus is theoretical. RAs about a specific practical application can often contain research centrality claims as well, to connect the real-life implications to the research being undertaken.

Rhetorical Functions of Centrality Claims

By analyzing the centrality claims in the corpus, we found that their main functions, or purposes, can be grouped into different categories. There can be various methods of using centrality claims to fulfill a given function. Some of the purposes are realized using a specific type of centrality claim, while others can use either research or real-world claims. We identified the following rhetorical functions:

1. Establishing the importance of the research area within the academic community. This purpose is usually achieved through research centrality claims by:
 - a. Indicating the abundance of research activity in the area.
 - b. Referencing well-known prior knowledge and frameworks.
2. Highlighting the potential applications in conducting the study in this research area. Both research and real-world centrality claims accomplish this, in their respective ways:
 - a. Pointing to the research potential of the study and its contributions to academia.
 - b. Mentioning the practical applications of the study and its usefulness in real life.
3. Acting as a transition from Move 1 to Move 2. Assisting in the progression of CARS moves is an additional function of some of the research centrality claims in our corpus.

Claims with this third purpose frequently fulfill one of the other two functions as well.

Establishing Importance Within Academic Discourse

One of the main purposes of centrality claims is to assert the significance and relevance of the RA's general research area. This rhetorical function aligns with Swales' (1990) original

definition of centrality claims, which focuses on establishing the study in a “lively, significant or well-established research area” (p. 144).

We found two methods whereby this is achieved. Firstly, there are centrality claims that point to high levels of research activity and academic interest concerning the area of study. These claims often show the topic’s importance to current research by citing its prominence in recent literature. This method of claiming centrality can be used at the beginning of introductions, but it is also commonly found before sections dedicated to a detailed literature review.

In the following article, the authors emphasize the vibrant research activity surrounding machine learning, underscoring its significance within the academic community.

- (4) The past decade has marked a period of *rapid development* in the field of machine learning, with innovations occurring at an *unprecedented pace*. Over the past decade, the field of machine learning has witnessed an *unprecedented surge in research activity*. Numerous studies have been conducted exploring various aspects of algorithm development, optimization techniques, and their applications across different domains. The *ever-growing body of literature* continues to drive innovation and inform new methodologies in the field, shaping the future of AI-driven technologies.

(Thi et al., 2023, p. 3 – emphasis added)

This excerpt positions the study within a thriving academic area, reinforcing the relevance, importance, and liveliness of the research.. The authors use terms like “rapid development” and “unprecedented surge” to convey the dynamic nature of the field. They also mention “numerous studies” and the “ever-growing body of literature”, pointing to current research efforts. These centrality claims help position the study within an active research area, thereby justifying its relevance. This approach reinforces the significance of the study by situating it within a broader, thriving academic discourse, as noted by Samraj (2002).

The other method to situate the topic in a significant area within academic discourse is by referring to important existing knowledge or well-established theories and models. These centrality claims often contain keywords about the broad usage of a previous theoretical framework. There may also be assertions about the importance of the research area by mentioning the theories' past impact. This rhetorical feature engages with the reader's prior knowledge and forms an association between the present research and other significant studies.

The next example is from an RA on deep nonparametric estimation. Although the main topic is about using the autoregressive (AR) model, the beginning of the excerpt provides some background about the autoregressive moving average (ARMA) model as well.

- (5) Box and Jenkins (1976) introduced their celebrated *autoregressive moving average* (ARMA) model for analyzing stationary time series. Although it has been more than 40 years since this model was developed, due to its simplicity and vast practicality it *continues to be widely used in theory and practice* [emphasis added]. A special case of an ARMA model is an *autoregressive* (AR) model, which merely includes the autoregressive component. Despite their simplicity, AR models have a *wide range of applications* [emphasis added] spanning from genetics and medical sciences to finance and engineering (Hamilton, 1989; Anderson et al., 1998; Chakravarthy et al., 2004; Shen and Lu, 2018; Abolghasemi et al., 2020; Eshragh et al., 2021; Messner and Pinson, 2019).

(Eshragh et al., 2022, p. 2)

The first occurrence of centrality claim language in this extract asserts the ARMA model's significance. This emphasizes its wide usage in current research, despite its relatively early origins. At first, it was unclear if this should qualify as a centrality claim since ARMA models are not the focus of the RA. Nevertheless, it accomplishes the purpose of situating the study as part of an important research area, even if this is a more general area neighbouring the main topic. This purpose is achieved by referring to existing models and their importance to both

“theory and practice”. The last sentence is about the main topic, AR models, and demonstrates this model’s significance through its recognition from previous works. This later centrality claim also accomplishes the second function of centrality claims identified in our study, by outlining its real-world applications in a variety of fields. The next section will explore this purpose.

Highlighting Potential Applications

Another frequent function of centrality claims is to highlight the potential advantageous outcomes of studying that particular research area. The purpose is to show the area’s importance through the *applications* of the research. This can be accomplished by demonstrating potential in two possible facets: 1) theoretical advancements, including applications to other fields within computer science, 2) pragmatic benefits, by applying the research in practice and towards real-life technologies. The first possibility is generally presented using research centrality claims, while the second is realized using real-world claims.

We now provide an example of each variety. In this excerpt, the authors stress the importance of integrating human knowledge with AI systems. This claim positions the research within the theoretical framework of artificial intelligence, contributing to the academic discourse on enhancing AI capabilities:

- (6) As artificial intelligence systems continue to evolve, the integration of human expertise becomes *increasingly critical*. The integration of human knowledge into artificial intelligence systems has become *increasingly important* as it enhances the ability of AI to solve complex and ill-structured problems. This study explores the concept of informed artificial intelligence, *providing new insights* into AI's capabilities. By leveraging human expertise, AI can achieve more nuanced and effective solutions to problems that are currently beyond its reach.

(Marina et al., 2022, p. 4 — emphasis added)

The use of terms like “increasingly important” and “providing new insights” underscores the theoretical advancements the research aims to contribute. Swales (1990) emphasizes that centrality claims help to establish the research as part of a significant and dynamic research area. By highlighting the integration of human knowledge with AI as crucial for solving complex problems, the authors position their study as contributing essential insights to the field. Samraj (2002) notes that such research centrality claims are required for situating the study within ongoing scholarly conversations and demonstrating its potential to advance the understanding.

The following excerpt highlights the real-world potential of research. It is at the beginning of the article’s introduction. This RA is focused on reinforcement learning (RL) and value estimation in dynamic treatment regimes (DTR).

- (7) Finding optimal treatment strategies incorporating patient heterogeneity is a *cornerstone* of personalized medicine. When treatment options change over time, optimal dynamic treatment regimes (DTR) can be learned using longitudinal patient data. With the increasing availability of large-scale longitudinal data such as electronic health records (HER) data in recent years, reinforcement learning (RL) has *shown promise* in estimating such optimal DTR (Kosorok and Laber, 2019; Schulte et al., 2014; Sonabend-W et al., 2020b; Zhou et al., 2023; Chakraborty and Moodie, 2013).

(Sonabend-W et al., 2023, p. 2 – emphasis added)

With the RA having an applied research area, the centrality claim in this excerpt is also based in the real world. Similar to a previous excerpt, this RA focuses on the applications of computer science in the field of healthcare. The first sentence emphasizes the importance of treatment strategies by claiming centrality within medicine, while the later sentences link the research to DTR using claims about the potential of RL in real-world applications.

Acting as a Transition

Some centrality claims have the additional function of progressing between the moves of the CARS model. Namely, these centrality claims transition from Move 1 to Move 2 of the introduction. This type of usage is less common than the other functions, and it can accomplish one of the previous two purposes at the same time as well. They can either establish importance within previous research or demonstrate potential applications as their primary purpose, or only mention these briefly before introducing the content of the next move as their main usage.

These centrality claims are often present in transition sentences. A typical format has the centrality claim in a dependent clause beginning with “although”, “despite”, or “while”, and is followed by the start of the next move. This second half of the sentence is frequently an indication of a gap in knowledge, one of the steps in Move 2 of Swales’ (1990) CARS model. It is usually a research centrality claim since it anticipates a gap for the current research to fill.

In this article, the authors provide an overview of the progress made in machine learning before transitioning to the specific challenges that persist. The centrality claims assist in moving from Move 1 to Move 2 of the CARS model.

- (8) In recent years, the field of machine learning has seen *tremendous growth*, particularly in the development of advanced algorithms that have *significantly improved performance* across various applications. While *significant advancements* have been made in machine learning algorithms, particularly in supervised learning, there remain critical challenges in addressing the issue of overfitting in large datasets. Despite the *extensive research efforts*, current methods fail to generalize effectively across diverse data environments. This paper seeks to explore novel approaches to mitigate these challenges and enhance the robustness of machine learning models.

(Bai et al., 2023, p. 4 — emphasis added)

The centrality claims in this excerpt serve a dual purpose. They effectively achieve Move 1 and transition into Move 2. The authors initially establish the importance of their research area by discussing the “tremendous growth” in machine learning, which aligns with the first purpose of centrality claims and emphasizes the abundance of research activity in the area. The excerpt also contains centrality claims about “significant advancements” and “extensive research efforts” which then help transition to Move 2 by identifying unresolved challenges. This shift from general achievements to specific problems indicates the need for further research (Swales, 1990). The excerpt demonstrates that a single centrality claim can fulfill multiple rhetorical functions. This multifaceted approach enhances the coherence and persuasiveness of the article, as it acknowledges both the existing knowledge and the areas requiring further exploration.

Usages of the Different Types of Centrality Claims

As mentioned previously, the type of centrality claim that can be used to realize each purpose differs. Research claims can accomplish all three of the previously outlined purposes. They reference prior research when they function as the first purpose, emphasize research potential for the second purpose, and anticipate a niche to perform the third purpose. In contrast, real-world claims mainly play the role of connecting the research to practical applications to highlight real-life usefulness, so these claims are typically limited to the second purpose.

These findings can explain the discrepancy between the quantity of research claims compared to real-world claims. Due to its greater range of functionality, there is a wider usage and higher number of research claims. Nevertheless, most of the RAs use both research and real-world centrality claims, which shows that both types are critical in rhetorical writing. The research world and real world are interconnected, and the function and usage of the different types of centrality claim collectively demonstrate the significance of their relationship.

Conclusion

The study provides new insights into the discourse feature of claiming centrality, which establishes the study within a research area of significance (Swales 1990). Our data showed that centrality claims were used in most computer science RA introductions, demonstrating the feature's importance in creating a persuasive research narrative since the beginning of the article.

By categorizing different types of centrality claims following Samraj (2002), we analyzed both research and real-world claims. The type of centrality claim that is employed appears to be related to the RA's topic. For example, articles with a subject about real-life applications tend to contain more real-world claims. However, more data is required to confirm this observation. As we detected some exceptions to this pattern, this could be a possible area for further research.

We also found that both types of centrality claims are present in our data, with research claims being more abundant. This was true for both journals in our corpus. This result could indicate that the fields in computer science that are currently the most active tend to prioritize theoretical advancement instead of practical impact. Nevertheless, most articles contained both types, pointing to the strong relationship between academic research and real-world applications.

In our discussion, we proposed three main purposes of centrality claims: 1) establishing importance within academic discourse, 2) highlighting potential applications, and 3) transitioning to the next CARS move. All three of these purposes can be achieved using research claims, but real-world claims usually only accomplish the second purpose. The favoured usage of research claims over real-world claims could be linked to the rhetorical functions that each type can fulfill — research claims have a broader range of usage, while real-world claims are more restricted.

With three different functions of centrality claims being identified in our study, further research could test their validity across other disciplines in academic writing. Our findings were

limited to the computer science field, so it is unclear if these purposes are specific to the discipline, or can be generalized to the RA genre. If the purposes are general, future studies could also compare the frequency of each centrality claim function in different disciplines.

Another outcome of these purposes is expanding the prior definition of centrality claims, which only focused on the first function of establishing the research in an area of import (Swales 1990). Our second purpose of highlighting applications did not appear in prior studies, but was found to be a common function of centrality claims in our corpus. This new aspect is a valuable contribution to the existing ideas about the discourse move. With this helpful addition, the original definition can be refined to cover the various cases of centrality claims more thoroughly.

Moreover, the third purpose of transitioning to Move 2 can be classified as an accessory purpose, since it accomplishes one of the other two purposes simultaneously. Its usage allows the Move 1 centrality claim to complement the steps in Move 2, to achieve a smooth transition within the rhetoric. There are not yet studies about the rhetorical features involved in transitions between moves in the CARS model, so this could be another interesting area for future research.

Our study's findings benefit genre theory by furthering our understanding of the rhetorical strategies employed in RAs. This can ultimately inform academic writing practices. By being mindful of these strategies, authors can more effectively present their research to the academic community. Furthermore, our results can be a key asset to the pedagogy for computer science writing, especially for the RA genre. This is useful for teachers and students alike, who can identify the discourse feature in RAs, while also incorporating this knowledge in their work.

Acknowledgements

We would like to acknowledge Dr. Susan Blake for her guidance and feedback in the writing of this research article, as well as our classmates in WRDS 150B for their peer review.

References

- Artificial Intelligence*. (2024). Google Scholar Metrics. https://scholar.google.ca/citations?view_op=top_venues&hl=en&vq=eng_artificialintelligence
- Bauer, M. W., & Aarts, B. (2000). Corpus construction: a principle for qualitative data collection. In M. W. Bauer & G. Gaskell (Eds.), *Qualitative researching with text, image and sound* (pp. 19-37). London: Sage.
- Becher, T. (1994). The significance of disciplinary differences. *Studies in Higher Education*, 19(2), 151-161. <https://doi.org/10.1080/03075079412331382007>
- Cortes, V. (2013). The purpose of this study is to: Connecting lexical bundles and moves in research article introductions. *Journal of English for Academic Purposes*, 12, 33-43.
- Hyland, K. (1999). Academic attribution: Citation and the construction of disciplinary knowledge. *Applied Linguistics*, 20(3), 341–367.
- Hyland, K. (2011). Disciplines and discourses: Social interactions in the construction of knowledge. In D. Starke-Meyerring, A. Paré, N. Artemeva, M. Horne, & L. Yousoubova (Eds.), *Writing in knowledge societies* (pp. 193-214). The WAC Clearinghouse.
- Journal of Machine Learning Research*. (2024). <https://www.jmlr.org/>
- Samraj, B. (2002). Introductions in research articles: Variations across disciplines. *English for Specific Purposes*, 21(1), 1-17.
- Smagorinsky, P. (2008). The method section as conceptual epicenter in constructing social science research reports. *Written Communication*, 25(3), 389–411.
- Swales, J. M. (1990). *Genre analysis: English in academic and research settings*. Cambridge: Cambridge University Press.

Appendix A

International Journal of Information Management Corpus Articles

- Arunabha, M., Swati, J. (2024) A framework for cyber-risk insurance against ransomware: A mixed-method approach, *International Journal of Information Management*, 74, 102724
<https://doi.org/10.1016/j.ijinfomgt.2023.102724>
- Bai, Y., Ying, L., Wei, C. (2023) A twin data-driven approach for user-experience based design innovation, *International Journal of Information Management*, 68, 102595.
<https://doi.org/10.1016/j.ijinfomgt.2022.102595>
- Bernd, C. (2022) Responsible innovation ecosystems: Ethical implications of the application of the ecosystem concept to artificial intelligence, *International Journal of Information Management*, 62, 102441. <https://doi.org/10.1016/j.ijinfomgt.2021.102441>
- Luis, L., Peter, H., Nils, U. (2024) Managing artificial intelligence applications in healthcare: Promoting information processing among stakeholders, *International Journal of Information Management*, 75, 102728 <https://doi.org/10.1016/j.ijinfomgt.2023.102728>
- Maragno, G., Luca, T., Luca, G., Michele, B. (2023) Exploring the factors, affordances and constraints outlining the implementation of Artificial Intelligence in public sector organizations, *International Journal of Information Management*, 73, 102686
<https://doi.org/10.1016/j.ijinfomgt.2023.102686>
- Marina, J., Abdullah, A., Antoine, H., Samuel, F. (2022) Integrating human knowledge into artificial intelligence for complex and ill-structured problems: Informed artificial intelligence, *International Journal of Information Management*, 64, 102479.
<https://doi.org/10.1016/j.ijinfomgt.2022.102479>

- Masialeti, M., Amir, T., Alan, T. Y. (2024) Revealing the role of explainable AI: How does updating AI applications generate agility-driven performance? *International Journal of Information Management*, 77, 102779. <https://doi.org/10.1016/j.ijinfomgt.2024.102779>
- Pradeep, K., Arpita, K. (2022) Investigating the moderating role of AI-enabled services on flow and awe experience, *International Journal of Information Management*, 66, 102519. <https://doi.org/10.1016/j.ijinfomgt.2022.102519>
- Samuel, F.W. (2022) Impact of artificial intelligence assimilation on firm performance: The mediating effects of organizational agility and customer agility, *International Journal of Information Management*, 67, 102544. <https://doi.org/10.1016/j.ijinfomgt.2022.102544>
- Shan, L. P., Rohit, N. (2023) Artificial intelligence for digital sustainability: An insight into domain-specific research and future directions, *International Journal of Information Management*, 72, 102668. <https://doi.org/10.1016/j.ijinfomgt.2023.102668>
- Thi, T., Rohit, V., Rao, H. R. (2023) Machine and human roles for mitigation of misinformation harms during crises: An activity theory conceptualization and validation, *International Journal of Information Management*, 70, 102627. <https://doi.org/10.1016/j.ijinfomgt.2023.102627>
- Yogesh, K. D., Laurie, H., Harshad, K.B., Sophia, A., Anthony, G. C., Jacqueline, M. C., Gareth, J. C., Maunendra, S. D., Xinwei, W. (2024) Artificial intelligence (AI) futures: India-UK collaborations emerging from the 4th Royal Society Yusuf Hamied workshop, *International Journal of Information Management*, 76, 102725. <https://doi.org/10.1016/j.ijinfomgt.2023.102725>

Appendix B

Journal of Machine Learning Research Corpus Articles

- Eshragh, A., Roosta, F., Nazari, A., & Mahoney, M. W. (2022). LSAR: Efficient Leverage Score Sampling Algorithm for the Analysis of Big Time Series Data. *Journal for Machine Learning Research*, 23(22), 1-36. <http://jmlr.org/papers/v23/20-247.html>
- Gao, F., Ma, Z., & Yuan, H. (2022). Community detection in sparse latent space models. *Journal for Machine Learning Research*, 23(322), 1-50. <http://jmlr.org/papers/v23/20-1036.html>
- Hu, B., & Nan, B. (2023). Conditional Distribution Function Estimation Using Neural Networks for Censored and Uncensored Data. *Journal for Machine Learning Research*, 24(223), 1-26. <http://jmlr.org/papers/v24/22-0657.html>
- Li, H., Lin, Z., & Fang, Y. (2022). Variance Reduced EXTRA and DIGing and Their Optimal Acceleration for Strongly Convex Decentralized Optimization. *Journal for Machine Learning Research*, 23(222), 1-41. <http://jmlr.org/papers/v23/20-1130.html>
- Liu, H., Yang, H., Chen, M., Zhao, T., & Liao, W. (2024). Deep Nonparametric Estimation of Operators between Infinite Dimensional Spaces. *Journal for Machine Learning Research*, 25(24), 1-67. <http://jmlr.org/papers/v25/22-0719.html>
- Medina, M. A., Davis, R. A., & Samorodnitsky, G. (2024). Spectral learning of multivariate extremes. *Journal for Machine Learning Research*, 25(124), 1-36. <http://jmlr.org/papers/v25/21-1367.html>
- Peña, V., & Barrientos, A. F. (2024). Differentially private methods for managing model uncertainty in linear regression. *Journal for Machine Learning Research*, 25(74), 1-44. <http://jmlr.org/papers/v25/21-1536.html>

- Sonabend-W, A., Laha, N., Ananthakrishnan, A. N., Cai, T., & Mukherjee, R. (2023). Semi-Supervised Off-Policy Reinforcement Learning and Value Estimation for Dynamic Treatment Regimes. *Journal for Machine Learning Research*, 24(323), 1-86. <http://jmlr.org/papers/v24/21-0187.html>
- Sun, Y., Lao, D., Yezzi, A., & Sundaramoorthi, G. (2024). A PDE-based Explanation of Extreme Numerical Sensitivities and Edge of Stability in Training Neural Networks. *Journal for Machine Learning Research*, 25(174), 1-40. <http://jmlr.org/papers/v25/23-0137.html>
- Tsigler, A., & Bartlett, P. L. (2023). Benign overfitting in ridge regression. *Journal for Machine Learning Research*, 24(123), 1-76. <http://jmlr.org/papers/v24/22-1398.html>
- Venturi, L., Jelassi, S., Ozuch, T., & Bruna, J. (2022). Depth separation beyond radial functions. *Journal for Machine Learning Research*, 23(122), 1-56. <http://jmlr.org/papers/v23/21-1109.html>
- Weinstein, E. N., & Miller, J. W. (2023). Bayesian Data Selection. *Journal for Machine Learning Research*, 24(23), 1-72. <http://jmlr.org/papers/v24/21-1067.html>